

ECON 4538 - Problem Set 1 - Fall 2026

University of Minnesota

Due: September 21th 11:59pm

Instructions

Please answer the following questions using a Quarto/Jupyter notebook (either PDF or HTML format). For mathematical questions (LS questions), handwritten solution is accepted, provided it is clear to read and is properly merged into the notebook. For coding questions, please use Julia language. Submit the file in the respective Canvas assignment. Be clear and explain the steps you are taking, specially in the coding questions. **Also, include at the end a small section where you describe how you used AI tools to support you during this Problem Set.**

Question 1 (40 pts): LS Exercise 3.1

Question 2 (60 pts)

Consider the following problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_t \beta^t \ln c_t$$

subject to

$$k_{t+1} + c_t = Ak_t^\alpha + (1 - \delta)k_t \quad t = 0, 1, 2, \dots$$
$$k_0 > 0 \quad \text{given}$$

for $\beta \in (0, 1)$, $\alpha \in (0, 1)$, $\delta \in (0, 1]$ and $A > 0$.

does this need to be a list from 1-100

Part 1 (30 pts): Consider an uniform grid of size 100 in the k space $\{k_0, k_1, \dots, k_{99}\}$ where $k_0 = 0$ and $k_{99} = 10$. In Julia, program a function that receives as arguments $(\beta, A, \alpha, \delta)$ and its output should be (V, h) as a result of a value function iteration, where

- $V = \{V_0, \dots, V_{99}\}$ is a vector of size 100, representing the value function. $V_i = V(k_i)$.
- $h = \{h_0, \dots, h_{99}\}$ is a vector of size 100, representing the policy function (of the next period capital). $h_i = h(k_i)$.

Also make the function to report the iteration number for which the convergence was reached, and the time taken.

As a reference, consider the following steps for the function:

1. Start with a guess $V_0 = \{V_{0,0}, V_{0,1}, \dots, V_{0,99}\} = \{-100, 0, 0, \dots, 0\}$. (*why first element is set -100?*)
2. For iteration j , you have $V_j = \{V_{j,0}, V_{j,1}, \dots, V_{j,99}\}$. Compute $V_{j+1} = \{V_{j+1,0}, V_{j+1,1}, \dots, V_{j+1,99}\}$ as

$$V_{j+1,m} = \max_{n \in \{0,1,\dots,99\}} \ln(Ak_m^\alpha + (1 - \delta)k_m - k_n) + \beta V_{j,n}$$

for each $m \in \{0, 1, \dots, 99\}$. Impose $V_{j+1,0} = -100$ (*again, why?*).

3. Save $n^* = \{n_0^*, n_1^*, \dots, n_{99}^*\}$, solution indexes for the above problems.
4. Compute $d(V_{j+1}, V_j) = \max_{m \in \{0, \dots, 99\}} |V_{j+1,m} - V_{j,m}|$
5. If $d(V_{j+1}, V_j) > 10^{-7}$, proceed to iteration $j + 1$, starting from step 2. Otherwise, keep $V = V_{j+1}$.

6. Compute $h = \{k_{n_0}^*, k_{n_1}^*, \dots, k_{n_{99}}^*\}$

Part 2 (15 pts): For $\beta = 0.95$, $\alpha = 0.33$, $A = 1$ and $\delta = 1$, plot the resulting V against the grid of k . In the same graph, plot the analytical solution of V . Repeat this analytical vs numerical comparison with (1) the policy function h and (2) the consumption.

Part 3 (15 pts): Now change $\delta = 0.8$. Plot the resulting V , h and consumption. Compare them with the previous part results.

notes:

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PS1

Advanced Labor Economics

Problem One

$$\max E_0 \sum_{t=0}^{\infty} \ln c_t$$

$$\text{s.t. } c_t + k_{t+1} \leq A k_t^{\alpha} \theta_t$$

Value Iteration function

$$V(k, \theta) = \max_{k'} \ln(A k^{\alpha} \theta - k') + \beta E V(k', \theta')$$

$$\text{where } k' = h_1 A k^{\alpha} \theta$$

use objective function to solve policy iteration function

Exercise

Exercise 3.1 Howard's policy iteration algorithm

Consider the Brock-Mirman problem: to maximize

cobb-douglas
trans

$$E_0 \sum_{t=0}^{\infty} \beta^t \ln c_t,$$

subject to $c_t + k_{t+1} \leq A k_t^\alpha \theta_t$, k_0 given, $A > 0$, $1 > \alpha > 0$, where $\{\theta_t\}$ is an i.i.d. sequence with $\ln \theta_t$ distributed according to a normal distribution with mean zero and variance σ^2 .

Consider the following algorithm. Guess at a policy of the form $k_{t+1} = h_0(A k_t^\alpha \theta_t)$ for any constant $h_0 \in (0, 1)$. Then form

$$J_0(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln (A k_t^\alpha \theta_t - h_0 A k_t^\alpha \theta_t).$$

Next choose a new policy h_1 by maximizing

$$\ln (A k^\alpha \theta - k') + \beta E J_0(k', \theta'),$$

where $k' = h_1 A k^\alpha \theta$. Then form

$$J_1(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln (A k_t^\alpha \theta_t - h_1 A k_t^\alpha \theta_t).$$

Continue iterating on this scheme until successive h_j have converged.

Show that, for the present example, this algorithm converges to the optimal policy function in one step.

Given

* standard normal $\theta_t \sim N(0, 1)$

$$\max E_0 \sum_{t=0}^{\infty} \ln c_t$$

$$\text{s.t. } c_t + k_{t+1} \leq A k_t^\alpha \theta_t$$

$$V(k, \theta) = \max_{k'} \ln (A k^\alpha \theta - k') + \beta E V(k', \theta')$$

$$k' = h_0 (A k^\alpha \theta)$$

objective to solve as to

Problem

itr?

$$J_0(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln (A k_t^\alpha \theta_t - h_0 A k_t^\alpha \theta_t)$$

has functional
form that is
easier to
compute

$$E_0 \sum_{t=0}^{\infty} \beta^t \ln [(1 - h_0) A k_t^\alpha \theta_t]$$

this is an okay soln

$$\left[E_0 \sum_{t=0}^{\infty} \beta^t [\ln(1 - h_0) + \ln(A k_t^\alpha \theta_t)] \right]$$

alternatively

Create expression on k_0 as function of k_0

$$k_0 = k_0$$

$$k_1 = h_0 A k_0^\alpha \theta_0$$

$$k_2 = h_0 A k_1^\alpha \theta_1 = h_0 A (h_0 A k_0^\alpha \theta_0)^\alpha \theta_1 \Rightarrow \text{using previous two periods}$$

$$\Rightarrow h_0^{1+\alpha} A^{1+\alpha} \theta_0^\alpha \theta_1^\alpha k_0^{\alpha^2}$$

$$k_3 = h_0^{1+\alpha+\alpha^2} A^{1+\alpha+\alpha^2} \theta_0^\alpha \theta_1^\alpha \theta_2^\alpha k_0^{\alpha^3}$$

$$k_t = C(t) \cdot k_0^{\alpha^t}$$

↓
initial solution

$$J(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln((1-h_0) A \theta_t + k_t)$$

expression J
as its
arguments k_0

$$E_0 \sum_{t=0}^{\infty} \beta^t [\ln((1-h_0) A \theta_t) + \ln(k_t)]$$

ignore everything w/o k_t

$$+ E_0 \sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha)$$

$$+ E_0 \sum_{t=0}^{\infty} \beta^t \ln[C(t) k_0^{\alpha^t}]$$

$$+ E_0 \sum_{t=0}^{\infty} \beta^t \ln([C(t) + \ln[k_0^{\alpha^t}]])$$

$$+ \sum_{t=0}^{\infty} \beta^t \ln[k_0^{\alpha^t}]$$

$$+ \sum_{t=0}^{\infty} \beta^t \alpha^t \ln(k_0)$$

$$+ \sum_{t=0}^{\infty} \beta (\alpha)^t \ln(k_0)$$

$$+ \ln(k_0) \sum_{t=0}^{\infty} (\alpha \beta)^t$$

$$+ \ln(k_0) / (1 - \alpha \beta)$$

why?

* Choose h_1 by
maximizing in
question

h_1 appears on
policy function on
capital

Want to find k'

Martin missed
 α

Converges

$$J_0(k', \theta') = \dots + \ln(k') / (1 - \alpha \beta)$$

the F.O.C.

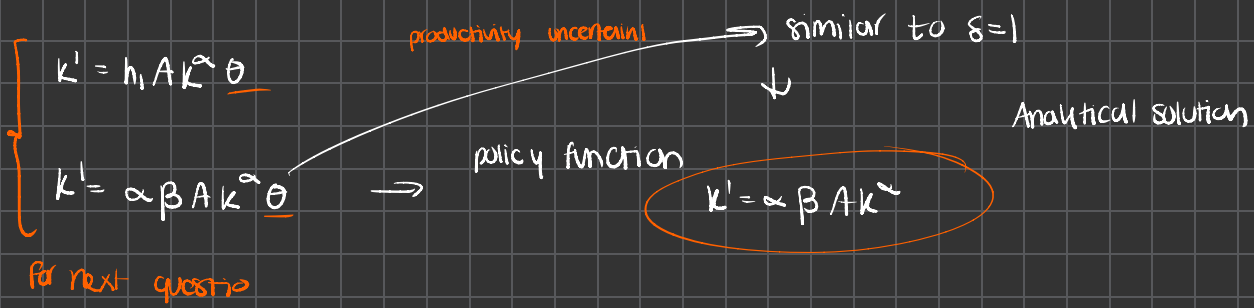
$$\frac{\partial J_0(k', \theta')}{\partial k'} = \frac{1}{1 - \alpha \beta} \cdot \frac{1}{k'}$$

we took arbitrary $h_0 \in (0, 1)$

$$\alpha \beta = h_1 \in (0, 1)$$

$$h_2 = \alpha \beta$$

the algorithm converge to the
same thing on the first step because we have a
Constant



$$\max_{h_1} \ln(Ak^\alpha \theta - h_1 Ak^\alpha \theta) + \beta \mathbb{E} J_0(k' = h_1 Ak^\alpha \theta, \theta')$$

Objective function
as h_1

$$\max_{h_1} \ln(Ak^\alpha \theta - h_1 Ak^\alpha \theta) + \beta \left(\dots + \frac{\ln(h_1 Ak^\alpha \theta)}{1-\alpha\beta} \right)$$

alternatively

$$\ln((1-h_1)Ak^\alpha \theta) + \beta \left(\dots + \ln h_1 Ak^\alpha \theta \right)$$

$$\ln(1-h_1) + \ln(Ak^\alpha \theta) + \beta \left(\dots + \left(\frac{\ln(h_1 Ak^\alpha \theta)}{1-\alpha\beta} \right) \right)$$

$$\downarrow$$

$$\frac{\ln(h_1) + \ln(Ak^\alpha \theta)}{1-\alpha\beta}$$

F.O.C.

$$0 = \frac{1}{1-h_1} + \beta / (1-\alpha\beta) \cdot \frac{1}{h_1}$$

Solve for h_1

$$\frac{1}{1-h_1} = \frac{\alpha\beta}{1-\alpha\beta} \cdot \frac{1}{h_1}$$

$$1-h_1 = \frac{1-\alpha\beta}{\beta} h_1$$

$$\alpha\beta - \beta h_1 = (1-\alpha\beta) h_1$$

$$\alpha\beta = \frac{(1-\alpha\beta + \beta)}{h_1}$$

$$\alpha\beta = 1 - \beta(1-\alpha) h_1 \Rightarrow$$

expression should
reduce to

$$\alpha\beta = h_1$$

$$\frac{\alpha\beta}{1+\beta(1-\alpha)} = h_1$$

Example of Policy Iteration - Bellman's Equation

Reinforcement learning

→ Policy iteration
→ Value iteration

Markov Decision Process

- Assume probabilistic state transition process

Goal: design policy that optimizes future reward

Assume environment evolves to markov decision process

↓
best policy is unknown
optimal state is unknown

$s \rightarrow s'$ transition to another state

state

Policy $\pi(s,a) = \Pr(a=a | s=s)$

Value

$V_{\pi}(s) = E[\sum_t \gamma^t r_t | s_0=s]$
value function
reward
discount rate

Solving policy function $\Rightarrow$ sequential to recursive problem w/ dynamic programming

exponential-decay reward
 $V_{\pi}(s) = E[\sum_k \beta^k r_k | s_0=s]$

V_{π} - policy function at state $s \Rightarrow V_{\pi}(s)$

E - expected future reward

$\beta \in [0,1]$
discount future reward to current reward
 c_t vs c_{t+1}

$V(s)$ - value of state s given best policy, π^*

Assume you make the optimal choice in $V(s)$

$V_{\pi}(s) = E[\sum_k \beta^k r_k | s_0=s]$

↓

$V(s) = \max_{\pi} E[\sum_{k=0}^{\infty} \beta^k r_k | s_0=s]$ sum of future rewards

Bellman's

↓

$V(s) = \max_{\pi} E(r_0 + \sum_{k=1}^{\infty} \beta^k r_k | s_1=s')$

next state after s'

dynamic $\Rightarrow$ function in current state to get to next best state

Value function reduces down to $\Rightarrow$ Given the policy function, what is the largest possible reward, or optimal policy I can receive

★ Optimize locally

$\pi^* = \arg \max_{\pi} E(r_0 + \beta V(s'))$

Policy that maximizes this value function

up down $\Rightarrow$ brute force

vs

bottom up

multistep optimization problem

Bellmans Equation $\Rightarrow$ know policy function $\Rightarrow$ value function $\Rightarrow$ sequential to recursive solution

Policy Function

Value Function

$$V_{\pi}(s) = \mathbb{E} \left[\sum_{k=0}^{\infty} \beta^k r_k \mid s_0 = s \right]$$

Bellmans

$$V(s) = \underset{\text{max}}{r_0} + \mathbb{E} \left[\sum_{k=1}^{\infty} \beta^k r_k \mid s_1 = s \right]$$

Value Iteration $\Rightarrow$ use bellmans equation

Consider

$$U(c_t) = \log c_t; \quad f(k_t) = k_t^\theta; \quad \delta = 1$$

Step ① Form Bellman Equation

Brock-Mirman $\delta = 1$

Law of Motion

$$c_t = f(k_t) + (1 - \delta)k_t - k_{t+1}$$

initial equation

$\Downarrow$

let $\delta = 1$ so $1 - \delta = 0$

$$c_t = f(k_t) - k_{t+1}$$

let $f(k_t) = k_t^\theta$

$\Downarrow$

$$c_t = k_t^\theta - k_{t+1}$$

Now use bellmans equation

$$V(k_t) = \max_{k_{t+1}} \left\{ \log(k_t^\theta - k_{t+1}) + \beta \mathbb{E} V(k_{t+1}) \right\}$$

inner outer ↑ reduces to -1

Step ② Derive F.O.C.

$$\frac{\partial V(k_t)}{\partial k_{t+1}} = \left\{ \frac{1}{k_t^\theta - k_{t+1}} (-1) + \beta \mathbb{E} V_{k_{t+1}}(k_{t+1}) \right\} = 0$$